Omni-Compass benchmark: Kubernetes native vs Kubernetes + Omni-Compass

Two identical kind clusters (1 control plane + 6 workers), same add-ons, workload (php-apache + HPA, target 50), load schedule and capture. **Native**: Omni-Compass not running. **Omni**: Omni-Compass drives the HPA target, the node pool (cordon/drain/uncordon), the power cap (in-place CPU limits on the app pods, enforced by the kernel), heat (harness law on live power), security (hold signal) and the rollout guard. Response times are real HTTP requests timed every 5 s. Power is a declared model (idle 100 W + 150 W x utilisation per worker in service; parked workers on standby at idle power), not a meter.

Gauges

Gauge	Native	Omni-Compass	Change
duration (min)	20.23	20.22	
worker nodes in service, mean	6.00	6.00	+0.0%
worker nodes in service, min	6.00	6.00	+0.0%
worker nodes in service, max	6.00	6.00	+0.0%
node-hours	2.02	2.02	-0.1%
power (W), mean	627	630	+0.5%
power (W), peak	652	669	+2.5%
energy (Wh)	211	212	+0.4%
energy, parked workers still on at idle power (Wh)	211	212	+0.4%
CPU used (cores), mean	0.829	0.918	+10.7%
CPU allocatable (cores), mean	24.00	24.00	+0.0%
utilisation (used / allocatable)	0.035	0.038	+10.7%
energy per core-hour (Wh)	756	686	-9.2%
node-hours per core-hour	7.23	6.53	-9.7%
pending pods, pod-minutes	0.500	0.000	-100.0%
pending pods, peak	2.00	0.000	-100.0%
HPA replicas, mean	8.86	8.98	+1.3%
HPA replicas, peak	10.00	10.00	+0.0%
HPA shortfall (desired > current), minutes	0.783	0.783	+0.0%
requests timed	1,065	1,145	+7.5%
time over the response line (% of samples)	0.094	0.000	-100.0%
response time (ms), mean	127	49.62	-61.0%
response time (ms), median	112	43.10	-61.4%
response time (ms), 95th percentile	275	72.40	-73.7%
response time (ms), 99th percentile	389	98.54	-74.6%
failed requests (%)	0.000	0.000	0

Significance (2-minute blocks: native 10, omni 10; bootstrap 95% interval)

Metric	Native	Omni-Compass	Change	95% CI of difference	Verdict
node_hours_per_core_hour	7.9647	7.5170	-5.6%	[-2.6585, +1.7548]	EVEN, no significant difference
kwh_per_core_hour	0.8284	0.7837	-5.4%	[-0.2508, +0.1696]	EVEN, no significant difference
utilisation	0.0342	0.0378	+10.4%	[-0.0069, +0.0148]	EVEN, no significant difference
pending_pod_minutes_per_hour	1.4400	0.0000	-100.0%	[-4.3200, +0.0000]	EVEN, no significant difference
hpa_shortfall_minutes_per_hour	2.2138	2.2680	+2.4%	[-3.6001, +3.8103]	EVEN, no significant difference

What Omni-Compass did

- decisions: 20 (one per minute)
- HPA target writes: 0
- node-pool resizes: 3
- scheduling-floor adds (pending pods): 0
- power-cap pod resizes (in place): 0
- rollout actions: 0
- power cap decided per minute: 1.0 1.0 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695
- heat (thermal) per minute: 0.358 0.392 0.42 0.445 0.467 0.486 0.502 0.517 0.53 0.541 0.551 0.557 0.562 0.567 0.572 0.576 0.579 0.583 0.585 0.586
- nodes decided per minute: 6 6 5 4 4 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3
- HPA target decided per minute (%): 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77

Reset

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restored target: 50
workers in service: 6 of 6
Omni records left after kill: none
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Minute by minute

min	nodes native	nodes omni	power W native	power W omni	replicas native	replicas omni	pending native	pending omni	CPU native	CPU omni
0	6.00	6.00	615	615	4.25	4.00	0.000	0.000	0.514	0.510

min	nodes native	nodes omni	power W native	power W omni	replicas native	replicas omni	pending native	pending omni	CPU native	CPU omni
1	6.00	6.00	616	616	5.00	5.50	0.000	0.000	0.542	0.573
2	6.00	6.00	615	617	5.00	6.00	0.000	0.000	0.505	0.563
3	6.00	6.00	624	618	5.00	6.00	0.000	0.000	0.665	0.612
4	6.00	6.00	632	636	8.50	8.50	0.500	0.000	0.973	1.07
5	6.00	6.00	632	633	10.00	10.00	0.000	0.000	0.958	0.986
6	6.00	6.00	628	635	10.00	10.00	0.000	0.000	0.903	1.06
7	6.00	6.00	642	655	10.00	10.00	0.000	0.000	1.23	1.59
8	6.00	6.00	647	649	10.00	10.00	0.000	0.000	1.38	1.43
9	6.00	6.00	645	656	10.00	10.00	0.000	0.000	1.30	1.62
10	6.00	6.00	637	654	10.00	10.00	0.000	0.000	1.10	1.56
11	6.00	6.00	618	615	10.00	10.00	0.000	0.000	0.586	0.514
12	6.00	6.00	617	616	10.00	10.00	0.000	0.000	0.558	0.526
13	6.00	6.00	621	624	10.00	10.00	0.000	0.000	0.688	0.731
14	6.00	6.00	631	632	10.00	10.00	0.000	0.000	0.955	0.992
15	6.00	6.00	630	635	10.00	10.00	0.000	0.000	0.919	1.04
16	6.00	6.00	630	632	10.00	10.00	0.000	0.000	0.950	0.986
17	6.00	6.00	623	627	10.00	10.00	0.000	0.000	0.740	0.834
18	6.00	6.00	616	620	10.00	10.00	0.000	0.000	0.515	0.653
19	6.00	6.00	617	615	10.00	10.00	0.000	0.000	0.571	0.485

Limits

- kind nodes are containers sharing one CI machine; timings and CPU are noisier than real servers.
- The two arms ran on two different CI machines at the same time; machine-to-machine variation is part of the noise.
- Energy is modelled from utilisation and nodes in service with the declared constants above.